Topic 7: Electric and Magnetic Fields

In order to develop their practical skills, students should be encouraged to carry out a range of practical experiments related to this topic. Possible experiments include using a coulomb meter to measure charge stored and using an electronic balance to measure the force between two charges.

Mathematical skills that could be developed in this topic include sketching relationships which are modelled by $y = k/x$, and $y = k/x^2$, using logarithmic plots to test exponential and power law variations, interpreting logarithmic plots and sketching relationships that are modelled by $y = e^{-x}$.

This topic may be studied using applications that relate to fields, for example, communications and display techniques.

Students should:
108. understand that an electric field (force field) is defined as a region where a charged particle experiences a force
109. understand that electric field strength is defined as $E = \frac{F}{Q}$ and be able to use this equation
110. be able to use the equation $F = \frac{Q_1 Q_2}{4\pi\varepsilon_0 r^2}$, for the force between two charges
111. be able to use the equation $E = \frac{Q}{4\pi\varepsilon_0 r^2}$ for the electric field due to a point charge
112. know and understand the relation between electric field and electric potential
113. be able to use the equation $E = \frac{V}{d}$ for an electric field between parallel plates
114. be able to use $V = \frac{Q}{4\pi\varepsilon_0 r}$ for a radial field
115. be able to draw and interpret diagrams using field lines and equipotentials to describe radial and uniform electric fields
116. understand that capacitance is defined as $C = \frac{Q}{V}$ and be able to use this equation
117. be able to use the equation $W = \frac{1}{2}QV$ for the energy stored by a capacitor, be able to derive the equation from the area under a graph of potential difference against charge stored and be able to derive and use the equations $W = \frac{1}{2}CV^2$ and $W = \frac{\frac{1}{2}Q^2}{C}$

Students should:
119. **CORE PRACTICAL 11: Use an oscilloscope or data logger to display and analyse the potential difference (p.d.) across a capacitor as it charges and discharges through a resistor.**
120. be able to use the equation $Q = Q_0 e^{-t/RC}$ and derive and use related equations for exponential discharge in a resistor-capacitor circuit, $I = I_0\, e^{-t/RC}$, and $V = V_0\, e^{-t/RC}$ and the corresponding log equations $\ln Q = \ln Q_0 - \frac{t}{RC}$, $\ln I = \ln I_0 - \frac{t}{RC}$ and $\ln V = \ln V_0 - \frac{t}{RC}$
121. understand and use the terms *magnetic flux density B*, *flux* ϕ and *flux linkage* $N\phi$
122. be able to use the equation $F = Bqv \sin\theta$ and apply Fleming's left-hand rule to charged particles moving in a magnetic field
123. be able to use the equation $F = BIl \sin\theta$ and apply Fleming's left-hand rule to current carrying conductors in a magnetic field
124. understand the factors affecting the e.m.f. induced in a coil when there is relative motion between the coil and a permanent magnet
125. understand the factors affecting the e.m.f. induced in a coil when there is a change of current in another coil linked with this coil
126. understand how to use Lenz's law to predict the direction of an induced e.m.f., and how the prediction relates to energy conservation
127. understand how to use Faraday's law to determine the magnitude of an induced e.m.f. and be able to use the equation that combines Faraday's and Lenz's laws $\mathscr{E} = \frac{-\mathrm{d}(N\phi)}{\mathrm{d}t}$
128. understand what is meant by the terms *frequency*, *period, peak value* and *root-mean-square value* when applied to alternating currents and potential differences
129. be able to use the equations $V_{rms} = \frac{V_0}{\sqrt{2}}$ and $I_{rms} = \frac{I_0}{\sqrt{2}}$